

MTD Design Projects — Exploring AI, Interaction & Humanity



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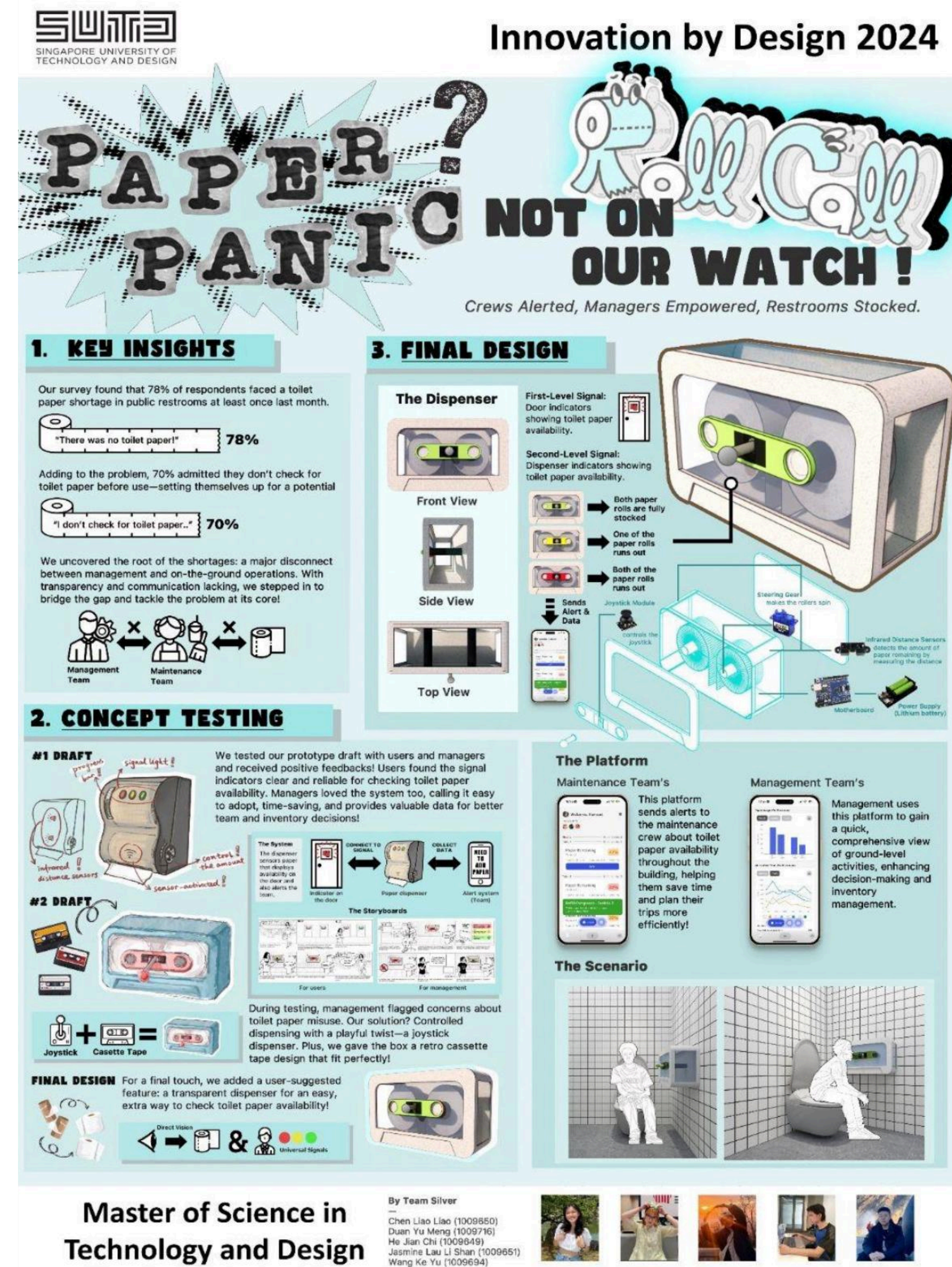
Project 1 : Smart Toilet Paper Detection & Dispenser System

Overview:

This project focuses on the design and application of a commercial smart toilet paper dispenser for public restrooms, malls, and hotels. AI and sensor systems enable automatic paper detection, intelligent dispensing, and visual feedback to improve facility management efficiency and user experience.

Contribution:

Responsible for internal circuit design and interaction logic, including remote dispensing control, lighting feedback, and distance-sensor-based detection.



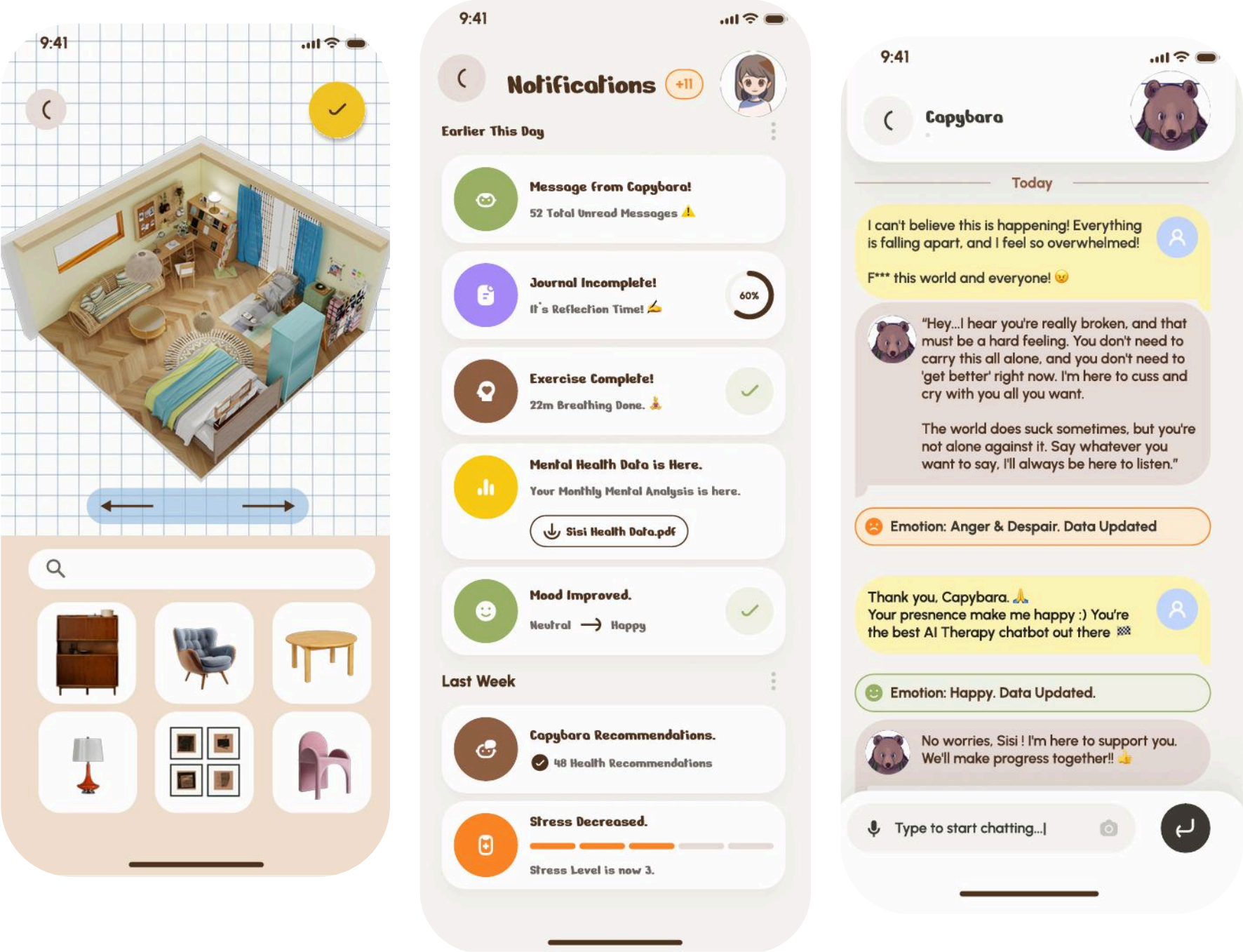
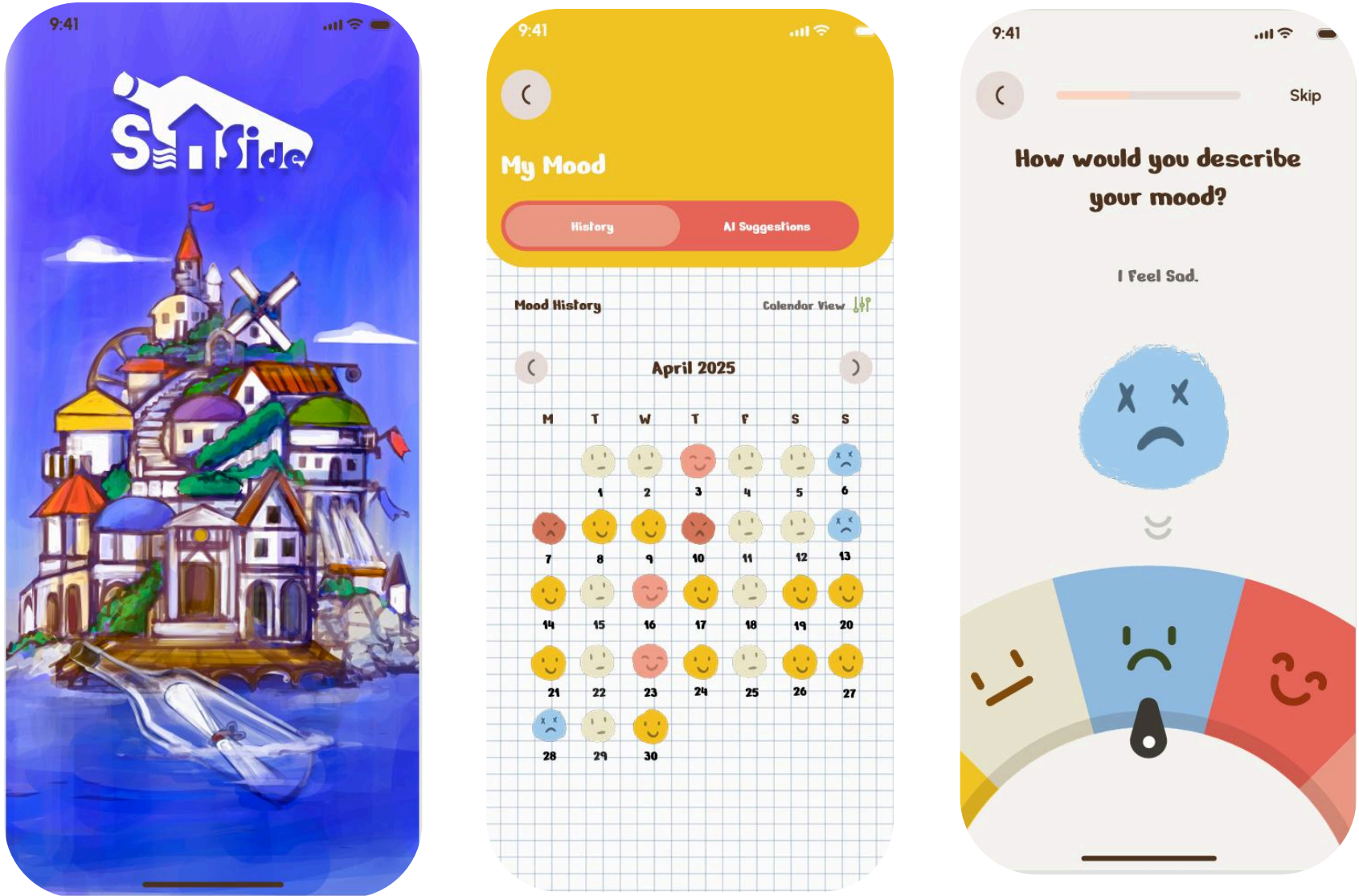
Project 2 : AI-driven Mental Health Support App for Teenagers

Overview:

This project leverages AI-based emotion recognition and semantic analysis to build a digital platform for teenagers' emotional awareness and support. The app enables safe, natural emotional expression and long-term well-being through anonymous interaction and visualization.

Contribution:

Responsible for app logic framework, interactive function design, and prototype development.



Core Feature Design of the App

1 Emotional "Message in a bottle " (Anonymous Emotional Release)

Provides an anonymous channel for emotional expression
Reduces social pressure
Theoretical Support – Anonymous Expression:
Minimizes external judgment and encourages genuine emotional disclosure

2 Diary Exchange (Long-Term Companionship and Peer Support)

Enables anonymous diary sharing between users
Fosters sustained supportive relationships and enhances psychological safety
Peer Support Theory: Promotes emotional healing through mutual understanding



3 AI Emotional Guidance (Positive Navigation)

Personalized mental health training
Actively guides users in emotional management
Transforms passive reception into proactive growth

4 Healing Space (DIY Personal Area)

Allows users to create a personalized healing space
Fulfills individual emotional needs
Provides users with a sense of emotional value and comfort

5 Emotion Visualization (Daily Mood Tracking)

Visual mood mapping and tracking
Serves as a tool for emotional self-awareness
Promotes the development of emotional intelligence

Project 3 : AI-Assisted Workflow for Heritage Architecture Restoration

Overview:

Research focusing on integrating AI into heritage restoration, emphasizing human-AI collaboration in digital reconstruction and structural prediction.

Contribution:

Applied existing AI models in architectural restoration scenarios, optimizing collaborative workflows between human experts and AI tools.

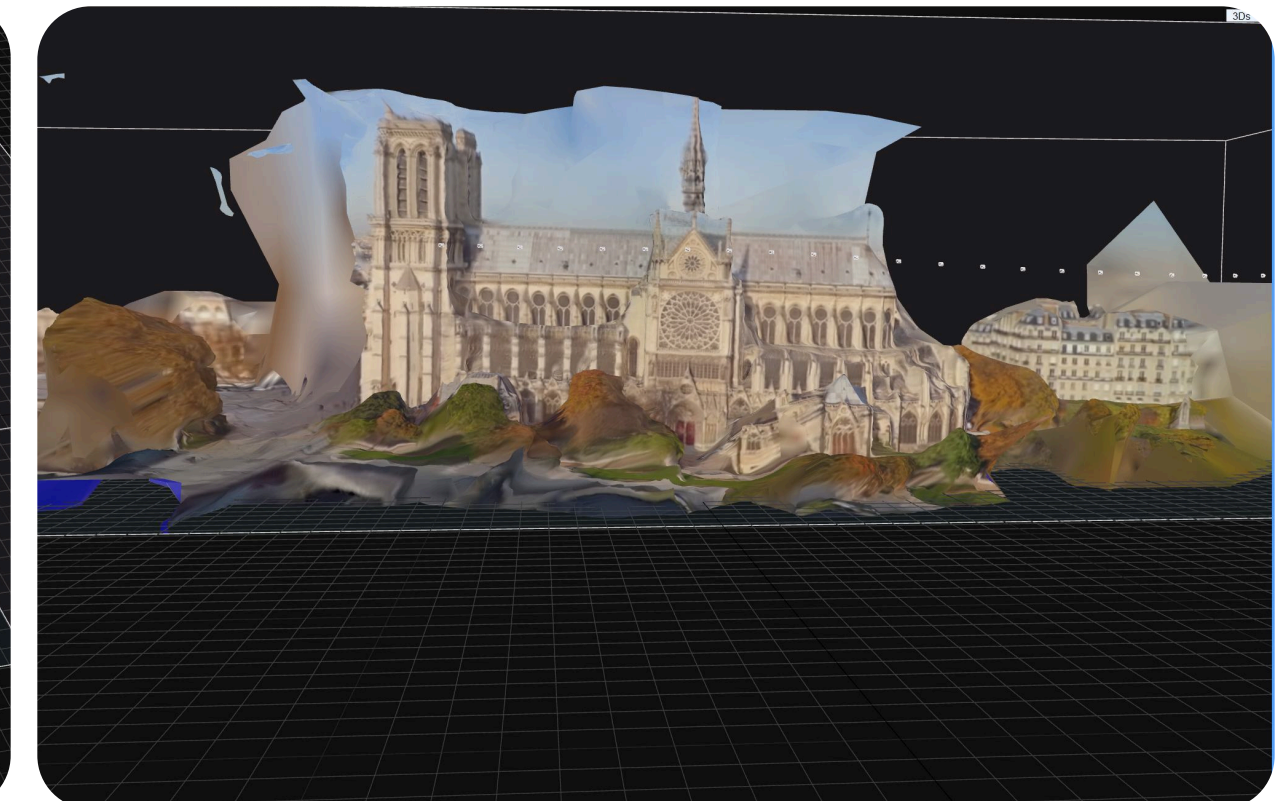
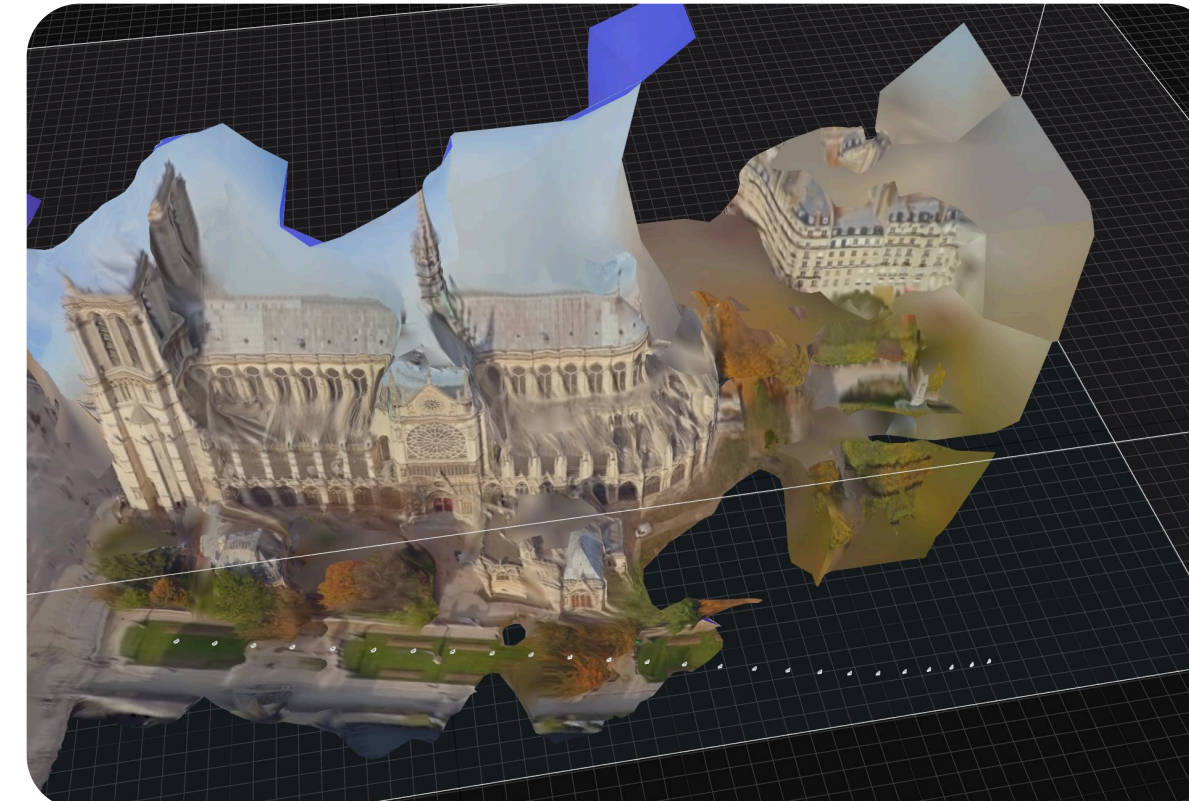
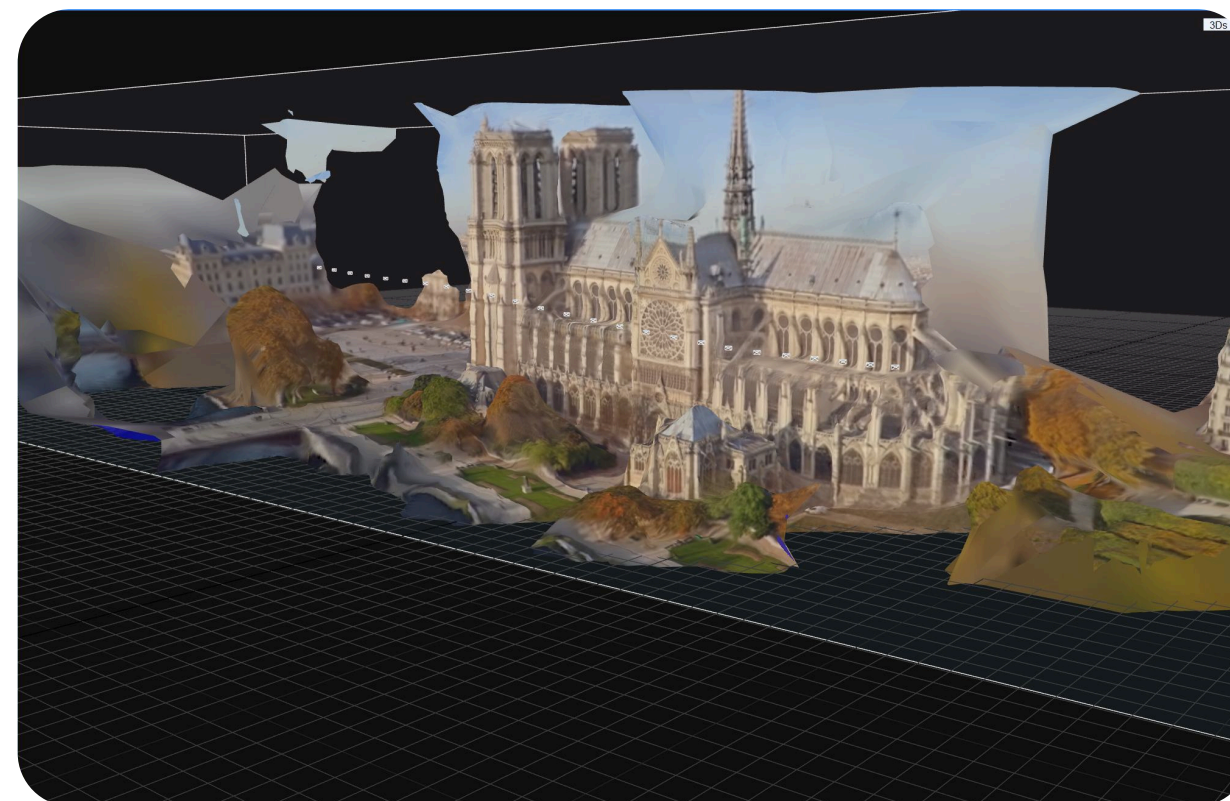
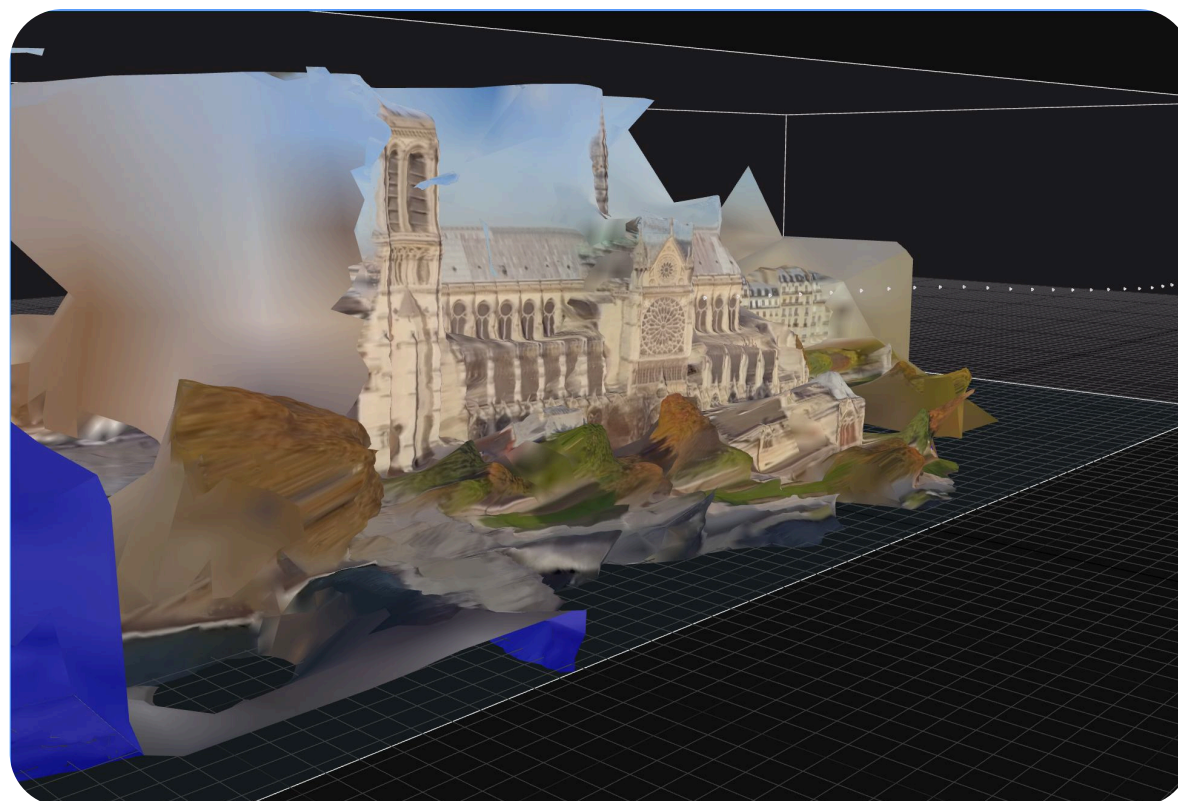
AR FOR ARCHITECTURAL HERITAGE CONSERVATION

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ABSTRACT

The preservation of architectural heritage is undergoing a transformative evolution driven by the integration of Artificial Intelligence (AI) and Augmented Reality (AR). Traditional restoration techniques, while valuable, struggle with issues such as information loss, time inefficiency, and limited scalability—especially when handling severely deteriorated or data-deficient cultural sites. This research proposes a novel, scalable, and participatory AI+AR-based framework for the restoration of architectural heritage. The project integrates deep learning tools, including Neural Radiance Fields (NeRF) and Generative Adversarial Networks (GANs), with human-centered validation and feedback loops involving experts and public contributors. Through a four-stage iterative methodology—baseline AI reconstruction, participatory validation, AI enhancement, and real-world AR deployment—the project enhances restoration accuracy, cultural authenticity, and public engagement. Our findings demonstrate significant improvements in restoration quality, reduced error rates, and increased stakeholder satisfaction, positioning this approach as a promising model for future digital heritage conservation initiatives.

Keywords: digital heritage;Augmented Reality ; artificial intelligence;



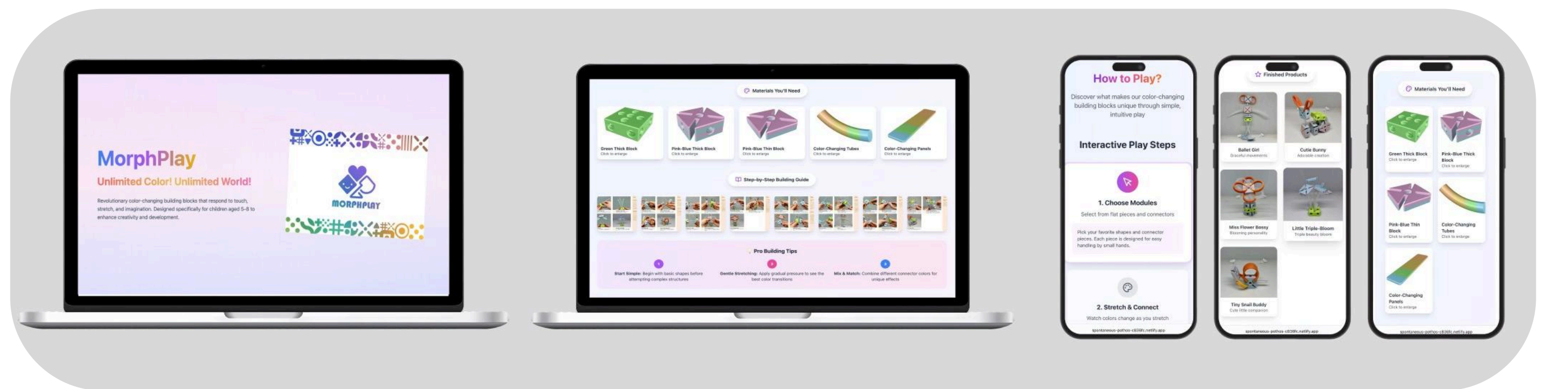
Project 4 : Color-Changing Building Blocks with AI-Enhanced Interaction App

Overview:

Combining color-changing materials with an AI-interactive app, the toy encourages creative building and sensory learning for children.

Contribution:

Responsible for design modeling, 3D printing prototyping, and structural iteration.



<https://spontaneous-pothos-c836fc.netlify.app/>

Project 5 : Trophy Design — When AI Meets Eastern Wisdom

Overview:

This philosophical design explores the convergence of Eastern wisdom and artificial intelligence, expressed through a symbolic, sculptural trophy form.

Contribution:

Responsible for 3D modeling and fabrication, contributing to symbolic expression and form development.



Research Reflection & Future Insight



Throughout my MTD journey, I have explored how AI can connect with human emotion, culture, and sensory experience. During this process, I have continually grown through teamwork, strengthening my interdisciplinary collaboration skills. In the future, I aim to further study tactile and empathetic interactions in intelligent product design—creating technologies that feel genuinely human.